Useful Inequalities for the Longest Run Distribution

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Abstract

General inequalities for the distribution of the longest success run in Bernoulli trials are found through a direct combinatorial analysis. Their employment allows the derivation of a new lower bound which can be used in the achievement of theoretical results.

Keywords: Bernoulli trials, longest success run, lower bound, asymptotic behavior.

1 Introduction

The study of success runs in Bernoulli trials has attracted the attention of several researchers both for its inherent theoretical interest and its possible applications. In particular Feller (1968) has introduced a new definition of run which allows its characterization as a recurrent event; through this approach he was able to analyze important properties of Bernoulli trials by applying general results from probability theory.

Great attention has been devoted to the distribution of the longest success run L_n in n Bernoulli trials whose range of application goes from the study of system reliability to the DNA sequence analysis. The exact expression for $R(n; k, p) = P(L_n < k)$, where p is the success probability and k is a non-negative integer, has been firstly presented in Burr and Cane (1961) and subsequently simplified by Lambiris and Papastavridis (1985), and Hwang (1986). They obtained a formula for $R(n; k, p)$ containing a single summation of ordinary binomial terms:

$$R(k;n,p) = \sum_{m=0}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^m p^{mk} q^{m-1} \left(\binom{n-mk}{m-1} + q \binom{n-mk}{m} \right) \tag{1}$$

where $q = 1 - p$ is the complementary failure probability and $\lfloor x \rfloor$ is the integer not greater than x. Nevertheless, the complexity of (1) may prevent its direct use in the analysis of theoretical properties of Bernoulli trials.

Several upper and lower bounds for $R(k; n, p)$ have been proposed to overcome this problem: a combinatorial analysis allowed Fu (1985) to achieve a first result in this direction

$$(1 - p^k)^{n-k+1} \leq R(k; n, p) \leq (1 - qp^k)^{n-k+1} \tag{2}$$

By following a different approach, Muselli (1997) obtained lower and upper bounds having similar expressions

$$(1 - p^k)^{n-k+1} \leq R(k; n, p) \leq (1 - p^k)^{\lfloor n/k \rfloor} \tag{3}$$

Unfortunately, the simplicity of these bounds is not accompanied by a small approximation error, which for some values of k, n, and p can be greater than 0.5.

The employment of the Poisson approximation, through the application of the Stein-Chen method (Chen, 1975; Arratia et al., 1989; Barbour et al., 1992), has produced a series of bounds with different complexity. However, their performance can be poor, especially for small n (Chao, 1995); the best results are obtained by using the following inequalities (Barbour et al., 1995)

$$\nu - \varepsilon \leq R(n, k; p) \leq \nu + \varepsilon \tag{4}$$

where

$$\begin{aligned} \nu &= e^{-qp^k(n-k+1)} - p^{k+1} e^{-qp^k(n-2k)} \\ \varepsilon &= \left(1 - e^{-qp^k(n-k+1)} + p^{k+1}\left(1 - e^{-qp^k(n-2k)}\right)\right)(2k+1)qp^k \end{aligned}$$

These bounds can often lead to smaller approximation errors than (2) and (3), but can also yield negative values for some choices of the variables k, n, and p.

Recently, new upper and lower bounds have been found (Muselli, 1998), which have the same form of (3) and seem to perform better than other available expressions:

$$(1-p^k)^{\frac{(n-k)q}{(1-p^k)^k}+1} \leq R(k; n, p) \leq (1-p^k)^{\frac{(n-k)q}{1-p^k}+1} \tag{5}$$

Unfortunately, the lower bound in (5) holds only when $kq \geq p$ and its closeness to $R(k; n, p)$ degrades when p approaches 1. Consequently, it cannot be used to establish general theoretical properties of Bernoulli trials.

In this paper basic inequalities for the probability $R(k; n, p)$ will be obtained, which allow the achievement of new lower bounds for the distribution of the longest success run L_n (Section 2). Through the application of these inequalities an interesting problem of combinatorial probability is approached and solved in Section 3. It concerns the derivation of proper expressions bounding the probability that the longest run in a sequence of coin tosses is formed by heads.

2 Bounds for the distribution of L_n

Consider n i.i.d. Bernoulli trials having success probability p (with $0 \leq p \leq 1$) and denote with $q = 1 - p$ the complementary failure probability. Let L_n be the length of the longest success run obtained during the n trials; the following notation will be used throughout this paper

$$R(k; n, p) = P(L_n < k)\ , \qquad Q(k; n, p) = 1 - R(k; n, p) = P(L_n \geq k)$$

It can be easily shown that if $1 \leq k \leq n$ and $0 < p < 1$, the function $R(k; n, p)$ strictly decreases with n and increases with k. On the contrary, the probability $Q(k; n, p)$ monotonically increases with n and decreases with k.

The following theorem can be useful in a variety of situations and allows the achievement of new lower bound for $R(k; n, p)$.

Theorem 1 *If $1 < k < n$ and $0 < p < 1$, the following inequalities hold*

$$R(k; m, p)R(k; n - m + k - 1, p) < R(k; n, p) < R(k; m, p)R(k; n - m, p) \tag{6}$$

for every integer m such that $k \leq m < n$.

Proof. Let us introduce the events

$$E_{jl} = \{\text{there exists a run with at least } k \text{ successes in the outcomes } X_j, \ldots, X_l\}$$

for $1 \leq j \leq l \leq n$, having denoted with X_i the outcome of the ith Bernoulli trial. The probability of these events is by definition

$$P(E_{jl}) = Q(k; l - j + 1, p)$$

Thus, in the hypothesis $k \leq m < n$ we have $E_{1n} = E_{1m} \cup E_{m-k+2,n}$, for which

$$Q(k; n, p) = Q(k; m, p) + Q(k; n - m + k - 1, p) - P(E_{1m} \cap E_{m-k+2,n}) \tag{7}$$

Now, let us consider the $k-1$ outcomes $X_{m-k+2}, \ldots, X_m$ and define the following system of events:

$$\begin{aligned} D_0 &= \{X_{m-k+2} = X_{m-k+3} = \cdots = X_m = \mathsf{S}\} \quad , \quad D_{k-1} = \{X_m = \mathsf{F}\} \\ D_j &= \{X_{m-k+j+1} = \mathsf{F}, X_{m-k+j+2} = X_{m-k+j+3} = \cdots = X_m = \mathsf{S}\} \quad \text{for } j = 1, \ldots, k-2 \end{aligned}$$

having denoted with S a success and with F a failure. As one can note, the index j counts the number of outcomes in $X_{m-k+2}, \ldots, X_m$ that precede the last success run. For the independence of Bernoulli trials we have

$$P(D_0) = p^{k-1} \quad , \quad P(D_j) = p^{k-j-1} q \quad \text{for } j = 1, \ldots, k-1$$

Then, the probability $P(E_{1m} \cap E_{m-k+2,n})$ can be written as

$$\begin{aligned} P(E_{1m} \cap E_{m-k+2,n}) &= P(E_{m-k+2,n} \mid E_{1m}) P(E_{1m}) = \\ &= Q(k; m, p) \sum_{j=0}^{k-1} P(E_{m-k+2,n} \mid D_j) P(D_j \mid E_{1m}) \end{aligned} \tag{8}$$

since the last $n-m+k-1$ trials do not depend on the first $m-k+1$, but only on $X_{m-k+1}, \ldots, X_m$.

Now

$$P(E_{m-k+2,n} \mid D_0) = P(X_{m+1} = \mathsf{S}) + P(X_{m+1} = \mathsf{F}) P(E_{m+2,n})$$

hence, we obtain

$$\begin{aligned} P(E_{m-k+2,n} \mid D_j) &= P(X_{m+1} = \mathsf{S}) P(E_{m-k+2,n} \mid D_j, X_{m+1} = \mathsf{S}) + P(X_{m+1} = \mathsf{F}) P(E_{m+2,n}) < \\ &< P(E_{m-k+2,n} \mid D_0) \end{aligned} \tag{9}$$

As a matter of fact, $P(E_{m-k+2,n} \mid D_j, X_{m+1} = \mathsf{S})$ is strictly less than 1, since not every sequence of outcomes $X_{m-k+j+1}, \ldots, X_n$ that begins with $k-j$ successes contains a run of at least k successes (for example when $X_{m+2} = X_{m+3} = \cdots, X_n = \mathsf{F}$).

Then, if $1 \leq j \leq k-1$, we have

$$P(D_j \mid E_{1m}) = \frac{P(D_j \cap E_{1m})}{P(E_{1m})} = \frac{P(D_j) P(E_{1,m-k+j})}{P(E_{1m})} = P(D_j) \frac{Q(k; m-k+j, p)}{Q(k; m, p)} < P(D_j) \tag{10}$$

because $Q(k; n, p)$ strictly increases with n.

By substituting (9) and (10) in (8) we obtain

$$P(E_{1m} \cap E_{m-k+2,n}) = Q(k;m,p) \sum_{j=0}^{k-1} P(E_{m-k+2,n} \mid D_j) P(D_j \mid E_{1m}) =$$

$$= Q(k;m,p) \sum_{j=1}^{k-1} P(E_{m-k+2,n} \mid D_j) P(D_j) + Q(k;m,p) P(E_{m-k+2,n} \mid D_0) P(D_0 \mid E_{1m}) +$$

$$+ Q(k;m,p) \sum_{j=1}^{k-1} P(E_{m-k+2,n} \mid D_j) \left(P(D_j \mid E_{1m}) - P(D_j) \right) >$$

$$> Q(k;m,p) \sum_{j=1}^{k-1} P(E_{m-k+2,n} \mid D_j) P(D_j) +$$

$$+ Q(k;m,p) P(E_{m-k+2,n} \mid D_0) \left(\sum_{j=0}^{k-1} P(D_j \mid E_{1m}) - \sum_{j=1}^{k-1} P(D_j) \right) =$$

$$= Q(k;m,p) \sum_{j=0}^{k-1} P(E_{m-k+2,n} \mid D_j) P(D_j) = Q(k;m,p) Q(k;n-m+k-1,p) \tag{11}$$

being

$$\sum_{j=0}^{k-1} P(D_j \mid E_{1m}) = \frac{1}{P(E_{1m})} \sum_{j=0}^{k-1} P(D_j \cap E_{1m}) = 1$$

If we insert the lower bound (11) for $P(E_{1m} \cap E_{m-k+2,n})$ in (7) we have

$$Q(k;n,p) < Q(k;m,p) + Q(k;n-m+k-1,p) - Q(k;m,p) Q(k;n-m+k-1,p)$$

from which the lower bound in (6) for $R(k;n,p)$ directly follows.

On the other side, if we denote with $\overline{E_{jl}}$ the complement of the event E_{jl}, the upper bound for $R(k;n,p)$ can be easily found by noting that $\overline{E_{1n}}$ is strictly included in $\overline{E_{1m}} \cap \overline{E_{m+1,n}}$. From the independence of the events $\overline{E_{1m}}$ and $\overline{E_{m+1,n}}$ we obtain therefore

$$R(k;n,p) = P(\overline{E_{1n}}) < P(\overline{E_{1m}}) P(\overline{E_{m+1,n}}) = R(k;m,p) R(k;n-m,p) \qquad \blacksquare$$

The general validity of inequalities (6) is further emphasized by the following corollary, which can be verified by a direct inspection of the limiting cases.

Corollary 1 *If $0 \leq k \leq n$ and $0 \leq p \leq 1$, we have*

$$R(k;m,p) R(k;n-m+k-1,p) \leq R(k;n,p) \leq R(k;m,p) R(k;n-m,p) \tag{12}$$

for every integer m such that $k \leq m \leq n$. The left inequality is also valid for $0 \leq m < k$.

Theorem 1 and its corollary can be directly employed to reobtain bounds (3). It is sufficient to choose $m = k$ in (12); since (1) gives $R(k;k,p) = 1 - p^k$, we obtain

$$(1 - p^k) R(k;n-1,p) \leq R(k;n,p) \leq (1 - p^k) R(k;n-k,p)$$

and by iterating the application of (12) with $m = k$ inequalities (3) follow.

A slightly more involved application of Theorem 1 also allows us to achieve a new lower bound for $R(k;n,p)$.

Theorem 2 *The inequality*

$$R(k;n,p) \geq (1-p^k)^{n-k+1-l(k,p)(h(k,p)-1)} \tag{13}$$

where

$$l(k,p) = \left\lfloor \frac{n-k}{h(k,p)+1} \right\rfloor \quad , \quad h(k,p) = \left\lfloor \frac{1-p^k}{q} \right\rfloor \tag{14}$$

is satisfied for every $0 < p < 1$ *and* $1 \leq k \leq n - h(k,p)$.

Proof. By setting $m = k + h(k,p)$ in the lower bound of Corollary 1 we obtain

$$R(k;n,p) \geq R\left(k;k+h(k,p),p\right) R\left(k;n-h(k,p),p\right) \tag{15}$$

Now, when $k \leq n \leq 2k$ the straightforward application of the exact formula (1) gives

$$R(k;n,p) = 1 - p^k(1 + q(n-k)) \tag{16}$$

Since $1 \leq h(k,p) \leq k$ we have

$$R\left(k;k+h(k,p),p\right) = 1 - p^k \left(1 + q \left\lfloor \frac{1-p^k}{q} \right\rfloor\right) \geq 1 - p^k(2-p^k) = (1-p^k)^2$$

By iterating $l(k,p)$ times the application of (15) we obtain

$$R(k;n,p) \geq (1-p^k)^{2l(k,p)} R(k;n-l(k,p)(h(k,p)+1),p) \tag{17}$$

hence, by using the lower bound in (2) for $R(k;n-l(k,p)(h(k,p)+1),p)$ it follows that

$$R(k;n,p) \geq (1-p^k)^{2l(k,p)}(1-p^k)^{n-k+1-l(k,p)(h(k,p)+1)} = (1-p^k)^{n-k+1-l(k,p)(h(k,p)-1)}$$

The expression for $l(k,p)$ can be obtained by applying the condition

$$n - l(k,p)(h(k,p)+1) \geq k \quad \Leftrightarrow \quad l(k,p) \leq \frac{n-k}{h(k,p)+1} \quad \blacksquare$$

The lower bound of Theorem 2 has general validity and outperforms (or is equivalent to) the corresponding expression in (2) for any value of n, k, and p.

Corollary 2 *If* $l(k,p)$ *and* $h(k,p)$ *are given by (14) the following inequality holds for every* $0 < p < 1$ *and* $1 \leq k \leq n$

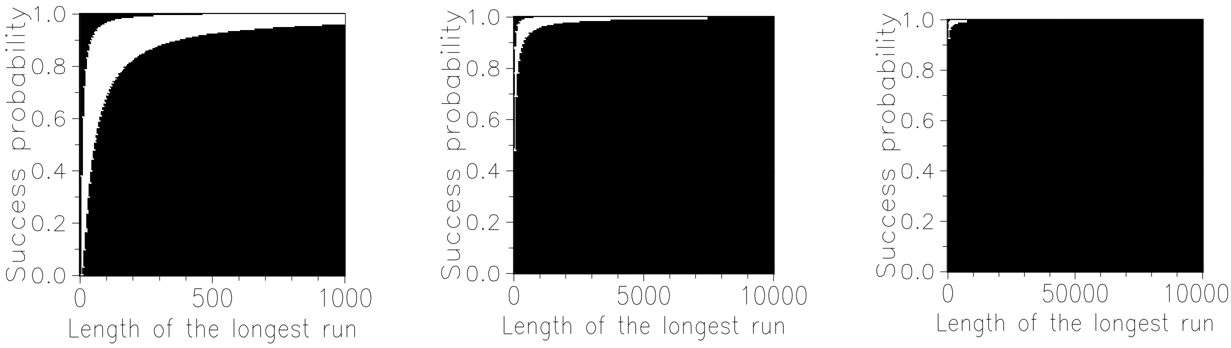
$$(1-p^k)^{n-k+1-l(k,p)(h(k,p)-1)} \geq (1-p^k)^{n-k+1}$$

Proof. It is sufficient to observe that $l(k,p) \geq 0$ and $h(k,p) \geq 1$. $\blacksquare$

A similar result for (4) is not immediate. Figure 1 shows for $n = 2,000$, $n = 20,000$, and $n = 200,000$ the region of the (k,p) domain where (13) gives a closer (or at least equivalent) approximation to $R(k;n,p)$ than lower bound in (4). The values of k greater than $n/2$ are not considered here, since in this region inequalities (4) are loose; on the other side, (16) gives a simple exact expression for $R(k;n,p)$ when $k \geq n/2$.

Large values of n have been considered for this comparison since the application of the finite Markov chain imbedding technique (Fu and Hu, 1987; Fu and Koutras, 1994) allows to reduce the computational cost needed to calculate the exact value of $R(k;n,p)$, when n is relatively small.

The following corollary provides a slightly different lower bound for the longest run distribution; it can be easily seen that it is better than (13) when $n-k$ is not divisible by $h(k,p)+1$.



a) b) c)

Figure 1: Region of the (k, p) domain where (13) gives a closer approximation to $R(k; n, p)$ than the lower bound in (4) for a) $n = 2,000$, b) $n = 20,000$, and c) $n = 200,000$.

Corollary 3 *The following inequality holds for every* $0 < p < 1$ *and* $1 \leq k \leq n$

$$(1 - p^k)^{2l'(k,p)} \leq R(k; n, p)$$

if $l'(k, p)$ *is defined as*

$$l'(k, p) = \left\lceil \frac{n - k + 1}{h(k, p) + 1} \right\rceil$$

being $\lceil x \rceil$ *the integer not less than* x.

Proof. In the expression (17) we can bound $R(k; n - l(k, p)(h(k, p) + 1), p)$ with $(1 - p^k)^2$ if

$$n - l(k, p)(h(k, p) + 1) \leq k + h(k, p) \qquad \Leftrightarrow \qquad l(k, p) \geq \frac{n - k + 1}{h(k, p) + 1} \qquad \blacksquare$$

From Corollary 3 a simpler (and slightly worse) lower bound for $R(k; n, p)$ can be directly obtained, which is more useful in the achievement of theoretical results. Section 3 will show an example of its application.

Corollary 4 *If* $\bar{R}(k; n, p)$ *is the upper bound in (5)*

$$\bar{R}(k; n, p) = (1 - p^k)^{\frac{(n-k)q}{1-p^k} + 1}$$

the following inequalities for $R(k; n, p)$ *hold*

$$\bar{R}^2(k; n, p) \leq R(k; n, p) \leq \bar{R}(k; n, p) \tag{18}$$

for $0 < p < 1$ *and* $1 \leq k \leq n - (1 - p^k)/q$.

Proof. The validity of the upper bound in (18) is proved in Muselli (1998). For the lower bound it is sufficient to apply Corollary 3, thus obtaining

$$R(k; n, p) \geq (1 - p^k)^{2l'(k,p)} \geq (1 - p^k)^{2\frac{n-k+h(k,p)+1}{h(k,p)+1}} = (1 - p^k)^{2\left(\frac{n-k}{h(k,p)+1} + 1\right)} \geq (1 - p^k)^{2\left(\frac{(n-k)q}{1-p^k} + 1\right)} \qquad \blacksquare$$

For the sake of completeness we report here a complementary result which compares the upper bound of (18) and that of (2). This allows us to conclude that inequalities

$$(1 - p^k)^{n-k+1-l(k,p)(h(k,p)-1)} \leq R(k; n, p) \leq (1 - p^k)^{\frac{(n-k)q}{1-p^k} + 1}$$

where $l(k, p)$ and $h(k, p)$ are given by (14), are a refinement of bounds (18) developed by Fu (1985).

Theorem 3 *For every* $0 < p < 1$ *and* $1 \le k \le n$ *we have*

$$(1-p^k)^{\frac{(n-k)q}{1-p^k}+1} < (1-qp^k)^{n-k+1}$$

Proof. It is sufficient to show that

$$(1-p^k)^{q/(1-p^k)} < 1-qp^k$$

or equivalently

$$q\log(1-x) < (1-x)\log(1-qx) \quad \text{for } 0 < x \le p$$

where we have set $x = p^k$.

The straghtforward analysis of the function

$$f(x) = (1-x)\log(1-qx) - q\log(1-x)$$

allows us to conclude that $f(x) > 0$ for every $0 < x \le p$, thus proving the assertion of the theorem. As a matter of fact, we have

$$f(0) = 0\ , \qquad f(p) = q\log(1-qp) - q\log q > 0$$

whereas the first derivative

$$f'(x) = -\log(1-qx) - q\frac{1-x}{1-qx} + q\frac{1}{1-x}$$

is positive in the interval $(0, p]$, since

$$f'(0) = 0\ , \qquad f'(p) = -\log(1-qp) + \frac{p}{1-qp} > 0$$

and the second derivative

$$f''(x) = q\frac{1}{1-qx} + qp\frac{1}{(1-qx)^2} + q\frac{1}{(1-x)^2}$$

is trivially positive in every point of its domain. ∎

3 Application to a problem of stochastic convergence

Theorem 1 and Corollary 4 can be directly used in the achievement of theoretical results for sequence of trials. An interesting problem which finds application in the convergence analysis of a training algorithm for neural networks (Muselli, 1997) is the following: consider two players A and B tossing a biased coin that gives a good outcome for A, e.g. head, with probability p and a good outcome for B, e.g. tail, with probability $q = 1 - p$.

If after n tosses of the coin, the player wins who has scored the longest sequence of favorable outcomes, we want to obtain proper bounds for the probability $V_n(p)$ that player A wins, that is the probability that the longest run is formed by successes (heads). Furthermore, is it possible to find the limit value for $V_n(p)$ when the number n of tosses increases indefinitely?

Let us begin our analysis by denoting with a_n (b_n) the length of the longest success (failure) run in the n trials; only one of the events $\{a_n > b_n\}$, $\{a_n < b_n\}$ and $\{a_n = b_n\}$ can occur. In the game between A and B these events correspond to {A wins}, {B wins} and {A and B tie} respectively; thus, we have $V_n(p) = P(a_n > b_n)$.

With these premises we can prove the following result:

Theorem 4 *The probability $V_n(p)$ that the longest run in n Bernoulli trials is formed by successes satisfies the following inequalities:*

$$V_n(p) \geq 1 - \sum_{k=0}^{n} R(k+1; n-2k, p)\,(R(k+1; n, q) - R(k; n, q)) \tag{19}$$

$$V_n(p) \leq \sum_{k=0}^{n} R(k; n-2k+1, q)\,(R(k+1; n, p) - R(k; n, p)) \tag{20}$$

Proof. Note that

$$V_n(p) = P(a_n > b_n) = \sum_{k=0}^{n} P(b_n < k \mid a_n = k) P(a_n = k) \tag{21}$$

and consider the following events:

$$\begin{aligned} F_{jl} &= \{\text{the length of the longest failure run in } X_j, \ldots, X_l \text{ is less than } k\} \\ G_i &= \{\text{the first longest run has } k \text{ successes and begins in } X_i\} \end{aligned}$$

for $1 \leq j \leq l \leq n$ and $1 \leq i \leq n-k+1$. Since $G_i \subset \{a_n = k\}$ we can write

$$P(b_n < k \mid a_n = k) = P(F_{1n} \mid a_n = k) = \sum_{i=1}^{n-k+1} P(F_{1n} \mid G_i) P(G_i \mid a_n = k) \tag{22}$$

But we have

$$P(F_{1n} \mid G_i) \leq P(F_{1,i-1} \mid X_{i-1} = \mathsf{F}) P(F_{i+k,n} \mid X_{i+k} = \mathsf{F}) \tag{23}$$

as a matter of fact the last $n-i-k+1$ outcomes (following the first run of k successes) do not depend on the first $i-1$ trials. Moreover, in all the sequences of n trials belonging to the event G_i we have that X_{i-1} and X_{i+k} are failures delimiting the first run of successes.

Now, since $F_{1,i-1} \subset F_{1,i-2}$ we obtain

$$\begin{aligned} P(F_{1,i-1} \mid X_{i-1} = \mathsf{F}) &= \frac{1}{q} P(F_{1,i-1}, X_{i-1} = \mathsf{F}) \leq \frac{1}{q} P(F_{1,i-2}, X_{i-1} = \mathsf{F}) = \\ &= P(F_{1,i-2}) = P(F_{1,i-1} \mid X_{i-1} = \mathsf{S}) \end{aligned}$$

and consequently

$$P(F_{1,i-1} \mid X_{i-1} = \mathsf{F}) \leq pP(F_{1,i-1} \mid X_{i-1} = \mathsf{S}) + qP(F_{1,i-1} \mid X_{i-1} = \mathsf{F}) = P(F_{1,i-1}) \tag{24}$$

A similar reasoning gives also

$$P(F_{i+k,n} \mid X_{i+k} = \mathsf{F}) \leq P(F_{i+k,n}) \tag{25}$$

But the probability of the events F_{jl} is by definition

$$P(F_{jl}) = R(k; l-j+1, q) \tag{26}$$

thus, by substituting (24), (25) and (26) in (23) we obtain

$$P(F_{1n} \mid G_i) \leq P(F_{1,i-1}) P(F_{i+k,n}) = R(k; i-1, q) R(k; n-i-k+1, q) \leq R(k; n-2k+1, q)$$

where we have used the first inequality in (12) with $m = i-1$.

Now, since the events G_i are disjoint and

$$\bigcup_{i=1}^{n-k+1} G_i = \{a_n = k\}$$

equation (22) becomes

$$P(b_n < k \mid a_n = k) \leq R(k; n-2k+1, q) \sum_{i=1}^{n-k+1} P(G_i \mid a_n = k) = R(k; n-2k+1, q)$$

The upper bound (20) follows from (21), being

$$P(a_n = k) = R(k+1; n, p) - R(k; n, p)$$

To show inequality (19) it is sufficient to repeat the same passages observing that

$$V_n(p) = P(a_n > b_n) = 1 - P(b_n \geq a_n) = 1 - \sum_{k=0}^{n} P(a_n < k+1 \mid b_n = k) P(b_n = k) \quad \blacksquare$$

The asymptotic values to which the probability $V_n(p)$ converges when the number n of trials increases indefinitely can be directly obtained by employing inequalities (18).

Theorem 5 *The probability $V_n(p)$ has the following limit values*

$$\lim_{n \to +\infty} V_n(p) = \begin{cases} 0 & \text{if } 0 \leq p < 1/2 \\ 1 & \text{if } 1/2 < p \leq 1 \end{cases}$$

Proof. Consider at first the case $0 \leq p < 1/2$; if $p = 0$ we have $R(k; n, 0) = 1$ for every $k \geq 0$. Consequently, (20) gives $V_n(0) = 0$.

In the general case $0 < p < 1/2$ let us break the summation at the right hand side of (20) in the two contributions corresponding to values of k preceding and successive to $h_n = \lfloor -\log n / \log \sqrt{pq} \rfloor$. Since $R(k; n-2k+1, q)$ increases with k, if $0 \leq k \leq h_n$ we obtain by applying the upper bound in (18)

$$\sum_{k=0}^{h_n} R(k; n-2k+1, q) \left(R(k+1; n, p) - R(k; n, p) \right) \leq R(h_n; n-2h_n+1, q) \leq$$
$$\leq \bar{R}(h_n; n-2h_n+1, q) \leq \exp\left(p(n-3h_n) \log\left(1 - q^{h_n}\right)\right) \tag{27}$$

In a similar way we have

$$\sum_{k=h_n+1}^{n} R(k; n-2k+1, q) \left(R(k+1; n, p) - R(k; n, p) \right) \leq 1 - R(h_n+1; n, p) \leq$$
$$\leq 1 - \bar{R}^2(h_n+1; n, p) \leq 1 - \exp\left((2n+2) \log\left(1 - p^{h_n}\right)\right) \tag{28}$$

Now, since $h_n = \lfloor -\log n / \log \sqrt{pq} \rfloor$, we obtain

$$q^{h_n} = \exp\left(\left\lfloor -\frac{\log n}{\log \sqrt{pq}} \right\rfloor \log q \right) \geq \exp\left(-\frac{\log q}{\log \sqrt{pq}} \log n \right) = n^{\alpha - 2} \tag{29}$$

$$p^{h_n} = \exp\left(\left\lfloor -\frac{\log n}{\log \sqrt{pq}} \right\rfloor \log p \right) \leq \exp\left(-\frac{\log p}{\log \sqrt{pq}} \log n - \log p \right) = \frac{n^{-\alpha}}{p} \tag{30}$$

where we have set

$$\alpha = \frac{\log p}{\log \sqrt{pq}} = \frac{2 \log p}{\log p + \log q}$$

It should be noted that $1 < \alpha < 2$ since $0 < p < \sqrt{pq} < 1$, for which, by substituting (29) and (30) in (27) and (28), we have

$$V_n(p) \le \exp\left(p\left(n - \frac{3\log n}{\log \sqrt{pq}}\right)\log\left(1 - n^{\alpha-2}\right)\right) + 1 - \exp\left((2n+2)\log\left(1 - \frac{n^{-\alpha}}{p}\right)\right) \tag{31}$$

But

$$\lim_{n\to+\infty} p\left(n - \frac{3\log n}{\log\sqrt{pq}}\right)\log\left(1 - n^{\alpha-2}\right) = \lim_{n\to+\infty} -pn^{\alpha-1} = -\infty$$
$$\lim_{n\to+\infty} (2n+2)\log\left(1 - \frac{n^{-\alpha}}{p}\right) = \lim_{n\to+\infty} -\frac{2n^{1-\alpha}}{p} = 0$$

being $\alpha > 1$ and $\log\sqrt{pq} < 0$. Thus, from (31) follows

$$\lim_{n\to+\infty} V_n(p) = 0$$

since $V_n(p) \ge 0$ for every feasible value of n and p.

In the complementary case $1/2 < p \le 1$ it can be shown with an analogous procedure that

$$\lim_{n\to+\infty} V_n(p) = \lim_{n\to+\infty}\left(1 - \sum_{k=0}^{n} R(k+1; n-2k, p)\left(R(k+1; n, q) - R(k; n, q)\right)\right) = 1 \quad \blacksquare$$

Returning to the problem of the game between the player A and B, we can assert by virtue of theorem (5) that when the number of tosses increases indefinitely the player with the highest probability of good outcome wins surely, even if this probability is slightly greater than 0.5.

In the case $p = q = 1/2$ (unbiased coin) the two players have the same probability $V_n(1/2)$ of winning (because of the evident symmetry of the problem) with the upper bound $V_n(1/2) \le 1/2$.

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